Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 06-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Clinical Study of Pembrolizumab (+) Berahyaluronidase Alfa (MK-3475A) to Treat Newly-diagnosed Metastatic Non-small Cell Lung Cancer (MK-3475A-F84) **Short Title/ Acronym:** MK-3475A-F84 **Protocol Number:** 3475A-F84 / U1111-1306-1214 **NCT Number:** NCT06698042 **Posting ID:** 872663339818887016 **Sponsor/Company Name:** Merck **Investigational Product:** Pembrolizumab (+) Berahyaluronidase alfa (MK-3475A) / Hyaluronidase (subcutaneous pembrolizumab) **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Medical Monitor, Merck

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) and safety of subcutaneous pembrolizumab coformulated with hyaluronidase (MK-3475A) versus intravenous (IV) pembrolizumab in the first-line treatment of participants with metastatic Non-Small Cell Lung Cancer (NSCLC) with a PD-L1 Tumor Proportion Score (TPS) $\ge 50\%$. **Secondary Objectives:** To evaluate efficacy outcomes and additional pharmacokinetic parameters including Maximum Serum Concentration (Cmax) after the first dose, Trough Concentration (Ctrough) after the first dose, and AUC at steady state.

2.2 Background & Rationale Researchers are investigating new, more efficient ways to treat metastatic NSCLC (cancer that has spread to other parts of the body). While IV pembrolizumab is a standard of care immunotherapy, administering it into a vein can be time-consuming. Pembrolizumab (+) Berahyaluronidase alfa allows for subcutaneous (SC) administration under the skin, which has the potential to significantly reduce treatment time. The study goal is to compare what happens to pembrolizumab in a person's body over time (PK exposure) when it is given as an IV infusion versus an SC injection.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Intravenous Pembrolizumab) **Blinding:** Open-label **Randomization:** Randomized (Experimental vs. Active Comparator)

3.2 Planned Enrollment & Duration Target N: 160 participants **Number of Sites:** Multicenter / Global **Estimated Study Start:** Nov-2024 **Study Status:** Recruiting

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of squamous or nonsquamous non-small cell lung cancer (NSCLC). 3. Tumors must have high PD-L1 expression (Tumor Proportion Score [TPS] $\ge 50\%$). 4. Measurable disease as assessed by the local site investigator/ radiology.

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Received prior systemic anticancer therapy for metastatic NSCLC. 3. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 4. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 5. Active autoimmune disease that has required systemic treatment in the past 2 years. 6. History of (noninfectious) pneumonitis/ interstitial lung disease requiring steroids or currently active pneumonitis/interstitial lung disease. 7. Active infection requiring systemic therapy.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-3475A (Pembrolizumab + Berahyaluronidase alfa) administered every 6 weeks (Q6W) versus standard Pembrolizumab IV administered Q6W. **Route of Administration:** Subcutaneous (SC) injection (Investigational) vs. Intravenous (IV) infusion (Comparator). **Duration of Treatment:** Treatment continues until disease progression, unacceptable toxicity, or up to the maximum permitted study duration.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care for symptoms. **Prohibited:** Concurrent investigational agents, live-virus vaccines, or any systemic immunosuppressive therapy (e.g., high-dose corticosteroids) not indicated for AE management.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Imaging, PD-L1 Testing | | **Baseline** | Day 0 | Randomization, First Dose (SC or IV), PK baseline sampling | | **Treatment** | Every 6 Weeks | Investigational Product Dosing, Safety Labs, PK Sampling | | **Interim** | Q9W to Q12W | Tumor

Efficacy Assessments (RECIST 1.1), Disease Monitoring | | **End of Study** | 30 Days post-last dose | Final Safety Follow-up, Survival tracking |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Area Under the Curve (AUC) of Pembrolizumab Measured After the First Dose:** Blood samples collected pre-dose and at multiple timepoints post-dose will be used to determine AUC exposure. * **Trough Concentration (C_{trough}) of Pembrolizumab Measured at Steady State:** Blood samples collected pre-dose and at multiple timepoints post-dose will be used to determine C_{trough} at steady state.

7.2 Secondary & Exploratory Endpoints * **Maximum Serum Concentration (Cmax) of Pembrolizumab Measured After the First Dose** * **Trough Concentration (C_{trough}) of Pembrolizumab Measured After the First Dose** * **AUC of Pembrolizumab Measured at Steady State** * Objective Response Rate (ORR) per RECIST 1.1, Progression-Free Survival (PFS), Overall Survival (OS), Duration of Response (DOR), and Incidence of Adverse Events (Safety Profile).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous reporting from informed consent through 30 days post-treatment. Special monitoring focus on injection site reactions associated with SC delivery. **Serious Adverse Events (SAEs):** Investigational sites must report all SAEs within 24 hours of awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee will oversee patient safety and efficacy trends throughout the trial's lifespan.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses; Safety Set for all AE assessments; PK Set for all pharmacokinetic measures. **Sample Size Justification:** Powered to demonstrate pharmacokinetic non-inferiority (AUC and C_{trough}) between the SC and IV formulations with a targeted enrollment of 160 participants. **Handling of Missing Data:** Standard oncology methodology applying rules of censoring for time-to-event outcomes (OS, PFS) and treating missing efficacy data as non-response for ORR.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring for data verification, protocol compliance, and source data review. **Audit Trail:** Robust Electronic Data Capture (EDC) system with full audit trails on eCRF locking, query management, and data clarifications.

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11. Study Identifiers & Governance

Primary ID: MK-3475A-F84 **Registry Number:** NCT06698042 **Posting ID:** 872663339818887016 **Sponsor/Lead Organization:** Merck **Study Title:** A Clinical Study of Pembrolizumab (+) Berahyaluronidase Alfa (MK-3475A) to Treat Newly-diagnosed Metastatic Non-small Cell Lung Cancer (MK-3475A-F84) **Official Regulatory Title:** A Phase 3 Randomized, Open-label Clinical Study to Evaluate the Pharmacokinetics and Safety of Subcutaneous Pembrolizumab Coformulated With Hyaluronidase (MK-3475A) Versus Intravenous Pembrolizumab, in the First-line Treatment of Participants With Metastatic Non-small Cell Lung Cancer With PD-L1 TPS 50% or Greater.

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12. Clinical Framework (NSCLC Specific)

Disease Areas: Lung Cancer, Non-Small Cell Lung Cancer **Preferred UMLS Name:** Non-Small Cell Lung Carcinoma **Diagnosis Threshold:** PD-L1 Tumor Proportion Score (TPS) $\ge 50\%$ **Medical Methodology:** First-line immune checkpoint inhibition **Optimization Target:** Efficient subcutaneous delivery of anti-PD-1 therapy

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13. Study Procedures & Methodology

Management Pathway: Standard first-line clinical oncology pathway for metastatic NSCLC **Education Protocol (HCP):** Administration instructions for subcutaneous hyaluronidase coformulation and injection site AE management **Education Protocol (Patient):** IV vs SC modality expectations, therapy adherence, and reporting of immune-related adverse events (irAEs) **Quality Audit Mechanism:** Tracking of PK sampling timing compliance and blinded independent central review (BICR) of radiology

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14. Observation & Follow-Up Schedule

Total Duration: Follow-up until progressive disease, death, or end of trial limits **Onsite Visit Cadence:** Pre-dose on Day 1 of every 6-week cycle **Remote Monitoring Frequency:** Intermittent telehealth reviews as applicable per site capability **Checklist Review Interval:** Radiological assessments every 9-12 weeks per protocol

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]